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Microarray-Based Cancer Classification via Optimized Gene Selection Using Evolutionary Population Strategies

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ABSTRACT: In the enormous field of bioinformatics research, a large volume of genetic data has been generated. Higher throughput devices are made available at a lower cost, making the generate of huge volumetric data. Handling such numerous data has become an extremely challenging task for selecting the relevant disease-causing gene. The development of microarray technology provides a higher chance of cancer diagnosis, by enabling to measurement of the expression level of multiple genes at the same time. Selecting the relevant gene by using classifiers for the investigation of gene expression data is complicated. Proper identification of genes from the gene expression datasets plays a vital role in improving classification accuracy. In this study, the major task is identifying highly relevant genes from the gene expression data for cancer treatment. By using a modified meta-heuristic approach known as parallel lion optimization (PLOA) for selecting genes from microarray data that are able to classify various cancer sub-types with more accuracy. The experimental results depict that PLOA outperforms LOA and other well-known approaches, considering the five benchmark cancer gene expression datasets. It returns 99% classification accuracy for the dataset, namely Prostate, Lung, Leukemia, and Central Nervous System (CNS), for the top 200 genes. Prostate and Lymphoma dataset PLOA is 99.19% and 99.93% respectively. On evaluating the result with other algorithms, the higher level of accuracy in gene selection is achieved by the proposed algorithm.

KEYWORDS: Cancer diagnosis, Parallel Lion Optimization, Gene Expression data, Genetic Algorithm, Classification.

1 Introduction

The death rate due to cancer is increasing in current days. Technology is fast-growing; there is a requirement for the detection and proper treatment of cancer. Microarray technology is gaining importance, where the expression of genes can be used to measure and classify cancer cells.

The microarray examination is a process where the RNA is mined and converted to mRNA, and cDNA is produced [1]. Both mRNA and cDNA are kept in a DNA microarray such that individual cDNA gets connected to its corresponding mRNA. Further, the image processing procedures are applied to identify the genes [2]. The samples are very less, and as the features are more, identification is the most motivating task. The main research is identifying the genes. Various methodologies for training the samples are available. Neural networks and machine learning approaches have been proposed. The dominant features of the gene to be selected, which help in accurate classification. The process

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may be classified into two stages. They are Feature selection and classification. The block diagram below shows the general model.

There are many gene selection techniques available, like wrapper, filter. Microarray data may have relevant and irrelevant data, which indicates high dimensionality. Feature selection supports selecting features that help in classification. It is a methodology where the available data is grouped depending on the feature [3].

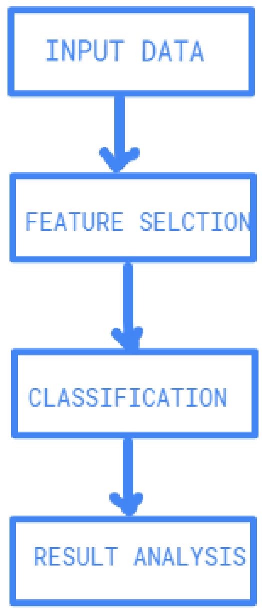

Figure 1: Block Diagram

Various classification techniques, which are already available, are used, like k-NN, convolutional neural networks, etc. Amongst the approaches, hybrid methodologies are gaining importance, which makes the system accurate. This study discusses on Parallel Lion Optimization method for efficient gene selection rate has been experimented with by comparing various techniques in previous approaches.

2 Literature Survey

The multi-objective PSO [4] repeatedly executes the steps of calculation of fitness, after which non-dominant sorting is done. GBest, which is a random solution among the results, is selected. This was mainly tested for colon cancer, Lymphoma, and leukemia datasets. Classification, namely Bayes family classifiers, function-based classifier, k-NN classifier, and tree-based classifier, was used. This was mainly concentrated on analyzing gene data. The lion algorithm identifies the best solution, which mainly performs the generation of pride, mating, territorial defense, and territorial takeover. The algorithm is defined based on the lion's social behavior.

Classification approach for micro data analysis [5] has two stages: feature selection, which is done with the help of embedded method, wrapper method, and filter method. Second are classifications for which neural networks and hybrid classifiers various as are used. There are other various approaches used, like gene classification, wavelet transformation, and swarm optimization, etc., where, depending on the feature, the technique can be implemented.

Biography-based optimization [6] is a kind of wrapper method in feature selection. It mainly works using the principle of special migration and mutation. This mainly avoids redundancy and gives optimal results. Redundancy-based filter works by listing the features in decreasing order and removing the related features from all features. IG-GA and dependency of feature are also the hybrid approaches that use branch and bound, sequential forward process, respectively, and provide optimal results.

Fuzzy approaches [7] are an order technique for feature selection. The fast algorithm initially removes irrelevant features and constructs the minimum spanning tree; it is partitioned based on the representative features. Bayes, C4.5 Ripper, and CFS classifier, along with this, give better results. There are various approaches for the detection of cancer using microarray data [8]. The simplified fuzzy ARTMAP, which uses cDNA microarray data for the identification of lymphoma. Gene expression graph represents the graph structure that relates silenced and expressed genes, where nodes are genes, and the edges represent gene expression relation [9]. Other approaches, like kernelized Fuzzy Rough set and semi-supervised SVM, which use biomarkers [10]. Transductive SYM adds the transductive samples to the original training set and continues the process of selecting and adding the transductive samples.

3 Proposed Methodology

3.1 Initialization

In the initial step, Npop arrangements are created arbitrarily in the seek space. %N of produced arrangements are arbitrarily picked as migrant lions. The rest of the populace will be haphazardly partitioned into P prides. Every arrangement in this calculation has a particular orientation of sex and remains steady amid the streamlining process. To imitate this reality, in every pride %S (%75–%90) of the whole populace shaped at the end are referred to as females and the remaining as males. For wanderer lions, this proportion is the other way about %(1_S). Lion is spoken to as taking after [11]:

$$\text{Lion} = [x_1, x_2, x_3, \ldots, x_{Nvar}] \tag{1}$$

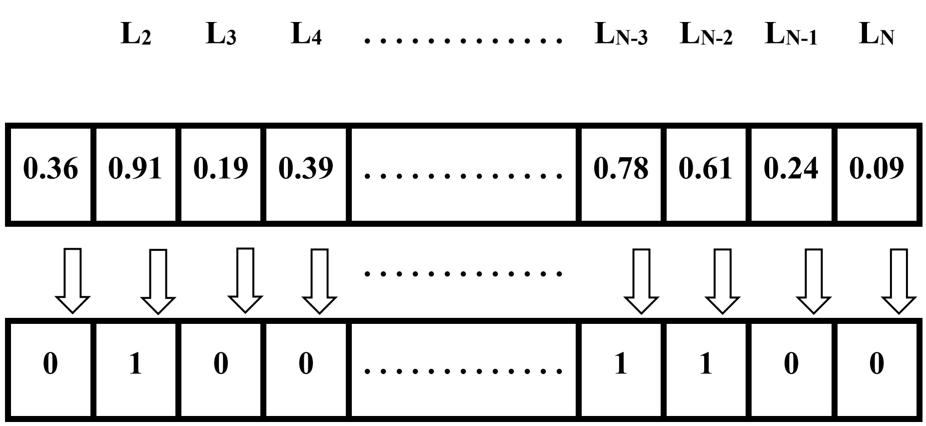

Figure 2: Lion Representation in PLOA

Figure 2 shows the gene representation of lion optimization. The integer higher than the maximum value is (0.50), and it will be 1, and the minimum value will be represented as 0. The genes set that have a value of 1 will be chosen for further optimization.

3.2 Hunting

In each pride, some females search for prey in a gathering to give sustenance to their pride. These seekers have specific methodologies to round up the prey and catch it. As a rule, lions took after similar examples when chasing. The lions are partitioned into seven distinctive stalking parts. PREY will escape from the seeker, and the new position of PREY is obtained as follows:

$$Hunter' = \begin{cases} \text{rand}\,((2 \times \text{PREY - Hunter}), \text{PREY}), (2 \times \text{PREY - Hunter}) < \text{PREY} \\ \text{rand}(\text{PREY}, (2 \times \text{PREY - Hunter})),\ (2 \times \text{PREY - Hunter}) > \text{PREY} \end{cases} \tag{2}$$

3.3 Moving Toward a Safe Place

As specified in the previous subdivision, in all pride a few females go chasing. Left out females go toward one of the regions. Since the region of every pride comprises the individual best so far, places of every part and helps the Lion Optimization Algorithm (LOA) to save the best arrangements acquired so far, after completing the course of emphasis, it can be utilized as significant and reliable data to enhance arrangements in LOA. The female lion's new position might be meant as

Female Lion’ = Female Lion + 2D × rand (0, 1) {R1} + U (−1,1) × tan (θ) × {R2} {R1}. {R2} = 0, ‖ {R2} ‖ 1 (3)

3.4 Roaming

Every male lion in a pride wander in that pride’s region because of a few reasons. To copy this conduct of occupant guys, the % R of pride in an area is chosen haphazardly and is governed by that lion. Along wandering, if an occupant male visits another position which is superior to anything its present best position, refresh its best went by arrangement. This wandering is a solid neighborhood inquiry and helps the Parallel Lion Optimization Algorithm (LOA) to look around for an answer to enhance it [12].

$$X \sim U\,(0, 2 \times d) \tag{4}$$

3.5 Mating Selection

Mating is a basic procedure that guarantees the lions’ survival and gives a chance for data exchange among individuals. In each pride, %Ma of is female lions mate with one or a few inhabitant lions. These guys are chosen haphazardly from an indistinguishable pride from the female to deliver posterity. For traveler lions, it’s diverse in that a wanderer female just mates with one of the males who are chosen by utilizing a roulette wheel determination process. The mating administrator is a direct blend of guardians for delivering two new offspring. As indicated by the accompanying conditions, the new offspring are created after choosing the female lion and male(s) for mating.

Roulette Wheel Selection

The thought behind the roulette wheel choice parent choice strategy is that every individual is allowed to end up a parent to the extent of their wellness assessment. It is called roulette wheel determination as the odds of choosing a parent can be viewed as turning a roulette wheel, with the extent of the opening for each parent being relative to its wellness. Clearly, those with the biggest wellness (and opening sizes) have more possibilities of being picked. Roulette wheel determination can be executed as takes after

a.: Sum the wellness of all the population individuals. Call this TF (add up to wellness).

b. Generate a random number m in the vicinity of 0 and TF.

c. Return the primary population part whose wellness added to the former populace individuals are more prominent than or equivalent to m.

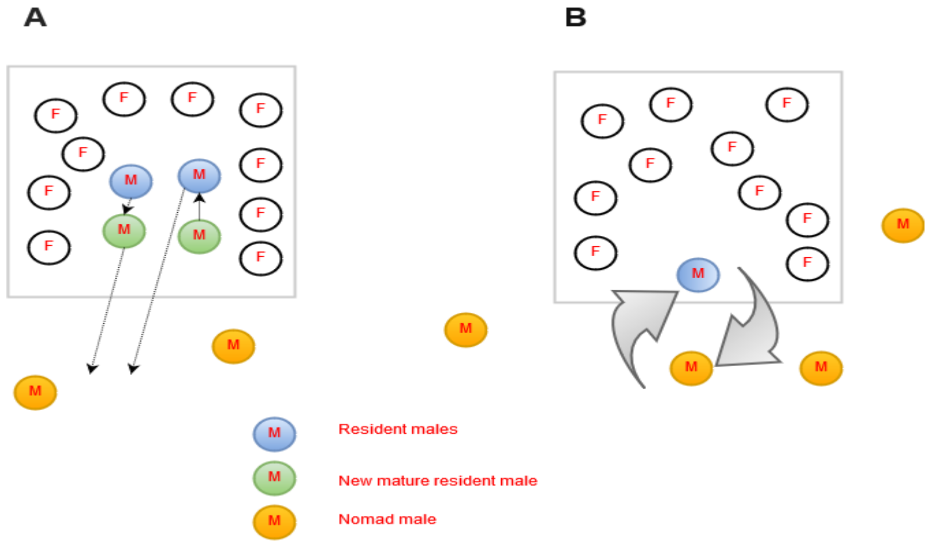

Figure 3: A and B representation of defense against new mature and no male

Figure 3 demonstrates the portrayal new developed occupant male choice. It is applied to the roulette wheel determination strategy rather than irregular choice. The issue with roulette wheel choice is that one part can overwhelm all the others and get chosen a great extent of times. The inverse is additionally valid. On the off chance that the calculation of the individuals will be close, then they will have a relatively measure up to possibility of being chosen [18]. To get around these issues, different wellness standardization methods can be applied.

$$Offspring_j \; 1 = \beta \times FemaleLion_j + \sum \frac{(1-\beta)}{\sum_{i=1}^{NR} S_i} \times MaleLion_j^i \times S_i \quad (5)$$

3.6 Defense

In a pride, male lions, when they develop, they end up forceful and fight other males in their pride. Beaten guys desert their pride and turn into wanderers. Then again, if a wandering male lion is sufficiently strong to endeavor to assume control over a pride by fighting its males, the beaten inhabitant male lion is driven out of the pride and becomes a traveler. The resistance administrator is partitioned into two fundamental advances: defense with respect to new mature resident males and defense with respect to nomadic males.

3.7 Migration

Roused by the lion's switch life and transitory conduct in nature when one lion sets out starting with one pride then onto another or switches its mode of life and the inhabitant female moves toward becoming a traveler and the alternative way around, it improves the genetic variety of the objective pride by its situation in the past pride. Then again, the lion’s movement and switch mode of life assemble the scaffold for data trade.

In each pride, the maximum number of females is prevented from being mined by S% of the population of the pride. For relocation administrators, some women chose arbitrarily and progress toward becoming wanderers. The size of the moved female in each pride is equivalent to surplus females in all prides, in addition to the maximum number of females in a pride. At the point when chosen females move from prides and move toward becoming migrants, new wanderer females and old traveler females are arranged by their fitness. At that point, the best females are chosen arbitrarily and dispersed to prides to fill the vacant place by moving females. This system keeps up the decent variety of the entire population and offers data among prides.

3.8 Convergence

For the halting condition, as normally considered in enhancement calculations, the best outcome is computed where the ceasing condition might be accepted as the CPU time, the greatest number of emphases, the number of cycles without change, and so forth.

The primary strides of the PLOA are condensed in the pseudo code. To perceive how PLOA can tackle enhancement issues, a few points might be noted: In PLOA, every arrangement has a specific sexual orientation, and every sex has its own particular system for looking. It helps PLOA to search for the ideal point by various methodologies [13]. The general point in utilizing a few prides is that each pride centers on a specific area and harmony between investigation and abuse. Its character of PLOA builds its ability to work for the enhancement of multi-modular issues. Traveler lions and their versatile wandering help PLOA to seek arrangement space haphazardly and escape from nearby optima. Individual best so far places of lions can give significant and solid data discovered so far by the populace. The proposed pride's region PLOA to made the best arrangements so far to finish the sequence of the cycle. In PLOA, by mating, lions share data between sexual orientations, while new whelps acquire character from the two sexes. Occupant guys wandering help PLOA to misuse data from their separate pride's neighbors that hold significant information. This method can be referred to as a solid neighborhood search. Specifically, the case ends up stale at a point. If best wellness esteem stagnates for a specific number of emphases in both the forests, then combine the whole population of lions in two timberlands and recently partition the prides and wanderers in light on their wellness value [14]. Figure 3 presents about the population relocation convergence in the dormant circumstance. The pseudo code portrays every progression in the Parallel Lion Optimization calculation amongst the two woodlands at the same time.

Pseudo code of PLOA

Begin

Define the fitness function $f(x),\ x = \{x_1, x_2, x_3, \ldots\ldots\ldots\ldots x_n\}$

Set the following parameters

Number of Lions, Number of Iterations, Number of Prides, Female Prides, Male Prides, Female Nomads, Male Nomads, Immigration Rate, Mating Probability, Numbers of lions for mating and moving

Define two forests and in each forest do the following steps in parallel

1. Generate random population of Lions N_{pop}
2. Initialize prides and nomad lions within the population

a. Randomly select N% of initial population as nomad lions. Randomly divide the remained lions into P prides and form each pride's region

b. In each pride S% of entire population are assumed as female lions and the remaining lions as males.

c. In each nomad (100-S) % of complete population are identified as females and the left out as males.

3. For each pride do

a. Some female lions are randomly selected and go hunting

b. Each of remained female lion in pride go near one of the best selected position from territory

c. In pride group, for each resident male; R% of territory lions are randomly selected and checked.

d. Select Ma% of female lions from pride using Roulette Wheel Selection

e. Selected females in pride mate with one or several resident male. Do mutation with Mutation Probability Mu. New cubs become mature. Merge new off springs with prides

f. Weakest male drive out from pride and become nomad

End For

Select some strongest male lions from nomad and go hunting

4. For Nomad do

a. Nomad lion (Both male and female) moving randomly in search space

b. Select Ma% of females from nomad using Roulette Wheel Selection

c. Selected nomad females mate with one of the best nomad male

d. Do mating with Mating Probability Mu. New cubs become mature

e. Merge new off springs with prides

f. Selected prides randomly attacked by nomad male

End For

5. For each pride do

a. Selected females from Pride with Immigration rate I become nomad

End For

6. Do

a. First based on their fitness vale each gender of the nomad lions is sorted. After that, the best females among them are selected and scattered to prides filling empty places of migrated females.

b. With respect to maximum acceptable number of each gender, nomad lions with smallest fitness value will be detached.

Find the overall best objective value from the whole population of lions

If best fitness value stagnates for particular number of iterations in both the forests

Merge the entire population of lions in two forests and newly divide the prides and nomads based on their fitness value

End If

If termination condition is not satisfied, then go to step 3.

4 Analysis

4.1 Datasets

In this study, the cancer benchmark microarray datasets from Kent Ridge Biomedical Data Repository [15] (http://datam.i2r.a-star.edu.sg/datasets/krbd/) are used for of experimental study. For the evaluation of the proposed method, five binary-class microarray cancer datasets are taken. The details of the datasets are given in Table 1. The third and fourth column representing Class1 and Class2, respectively, specifies the count of samples denoted inside parentheses. To evaluate our proposed algorithm, accuracy as a fitness function is used.

Table 1: Microarray gene expression datasets

Name of the Dataset	Number of Genes	Class1	Class2	Total Samples
Acute myeloid leukemia – Acute lymphoblastic leukemia (AML-ALL)	7129	ALL (47)	AML (25)	72
Prostate	12600	Normal (59)	Tumor (77)	136
Colon Tumor	2000	Tumor (40)	Healthy (22)	62
Lung Harvard2	12533	Adenocarcinoma (ADCA) (150)	Mesothelioma (31)	181
Diffuse large B-cell lymphoma (DLBCL) Harvard	7129	DLBCL (58)	Follicular lymphoma (FL) (19)	77

As mentioned earlier, the T-statistics for all microarray data genes are calculated and ranked based on their values. In this study, PLOA is applied to select informative genes from the top-M ranked genes.

The kNN classifier is used as the classifier for the comparison of the performance of CSC. In this study, top-50, top-100, and top-200 genes are selected from the dataset by applying the T-statistic measure. Figure 4.1 shows PLOA convergence over 200 iterations on the Leukemia dataset with top 50, 100, and 200 genes.

Figures 4 depict the accuracy obtained for selected top 50, 100, and 200 genes from T-statistics for leukemia, prostate, colon, lung, and lymphoma datasets. The results depict that the proposed PLOA algorithm outperforms existing statistical methods and PLOA in all five cancer gene expression data sets.

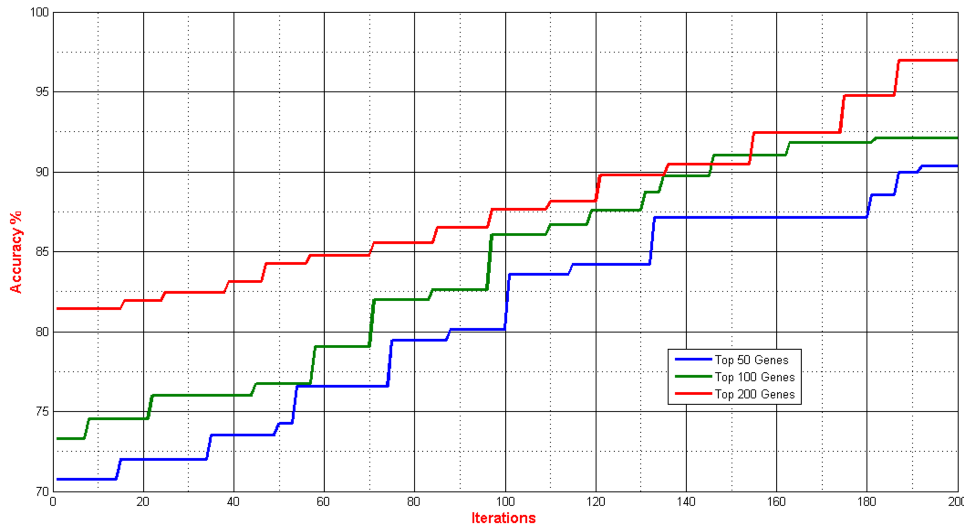

Figure 4: Convergence of the PLOA algorithm for the Leukemia Dataset

4.2 Comparative Analysis

We compared the efficiency of our approach with PSO, GA, and Harmony Search (HS). Table 2 depicts the maximum accuracy in a 200-epoch run of each method for each data set in the top 200 genes selection using the T-statistics measure. Seeing as the three benchmark methods are executing in the same experimental setup, inputs, and gene expression data. Therefore, the results are exactly accountable for comparison. The table of comparison shows that, compared to PSO, GA and HS have an obvious advantage. The proposed PLOA method results in 99% classification accuracy for Prostate, Lung Cancer, Michigan, and Lymphoma. For the Prostate dataset and Lymphoma outcome dataset, the classification accuracy obtained by the proposed classifier is 99.19% and 99.93% respectively. The results show that for most of the methods, the classification accuracy is dependent on the number of genes. When the number of genes increases, classification accuracy also increases. In terms of the correct rate, the search space of the solution capability of PLOA is better than the other three competitors.

Table 2: Comparison of our proposed algorithm with some important methods

Dataset	PSO	CS	HBMO	LOA	PLOA
Leukemia	83.51	90.34	88.55	92.22	96.12
Lymphoma	99.35	99.93	99.52	94.21	99.93
Lung	99.76	99.76	99.74	93.76	99.91

Prostate	88.50	87.66	88.37	91.45	99.19
Colon	96.64	97.86	98.81	95.23	99.94

4.3 Biological Analysis of Selected Genes for Leukemia Cancer Data

At last, the most significant subsets of genes were detected for every microarray data set. We included all subsets having the maximum accuracy, as well as listed the selected genes. Table 3 lists the highest selection frequency for the top 5 genes of the leukemia microarray data set. The roles or activities of the identified genes are confirmed by both the biological experiment as well as by the PLOA-based method. This indicates that it has a greater chance to act as biomarkers; these genes have been demonstrated to be the potential reason for leukemia cancer. Table 3 denotes the top 5 genes with the highest frequency in the leukemia data sets.

Table 3: Top 5 genes with the highest selection frequency of the leukemia data set

Dataset	PSO	GA	HS	LOA	PLOA
Leukemia	87.16	86.06	93.55	92.22	99.33
Lymphoma	91.77	93.86	88.56	94.21	99.80
Lung	90.33	93.87	96.62	93.76	99.19
Prostate	89.44	91.58	90.96	91.45	99.81
Colon	87.32	93.02	96.09	95.23	99.19

5 Conclusion

Microarray establishes the gene expression signatures associated with different phenotypes. Gene expression profiles are valuable to distinguish tumors into new and well-established tumor types. In this research, the meta-heuristic PLOA algorithm has been developed and implemented to select the informative genes from microarray data. An accuracy of classification is obtained by various eminent optimization algorithms like PSO, CS, HS, and LOA investigated. It is noticed that the genes selected by the PLOA for different cancer datasets have significant biological relevance. The result analysis acquired by the proposed algorithm is appropriate for the best gene selection, which has been evaluated with various gene expression benchmark datasets. The obtained accuracy by the PLOA method outperforms other methods.

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Conflicts of Interest: The authors declare no conflicts of interest to report regarding the present study.

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